

When Network Diversity Narrows at Sea: A Cross-Layer View of Internet Service Paths

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Abstract—Network-layer path diversity is often used to describe the alternatives available to an Internet service, but traceroute does not reveal whether those alternatives occupy distinct physical directions. We present CLASP, a paired cross-layer audit that constructs network-transition and feasible landing-region-corridor distributions from the same traceroute segments. We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements covering 13 DNS Roots, three applications, and two dynamic multi-target references. Under projection-score weighting, the Spearman rank association between network and corridor Top-2 concentration is 0.632 across 67 island and archipelagic service-facing units and 0.001 across 202 coastal-mainland units. The observed difference remains in the same direction under equal-share and Top-1 candidate allocation, where the corresponding gaps are 0.461 and 0.715. The relationship is heterogeneous within the service-facing population: 81 of 275 units move from a broad network distribution to a concentrated corridor distribution, while 158 remain broad at both layers. These measurements show that the correspondence between network and feasible physical-corridor diversity varies across geographic settings and cannot be inferred from network-layer breadth alone.

Index Terms—Internet measurement, submarine cables, traceroute, cross-layer analysis, anycast, path diversity

I. INTRODUCTION

Submarine cables carry nearly all intercontinental Internet traffic and connect users to remote services [1]. A cable or coastal corridor shared by several routes can expose those routes to the same physical failure. Evaluating path diversity therefore requires knowing whether routes that appear distinct at the network layer also occupy distinct physical directions.

Traceroute records routers and Autonomous System (AS) transitions, so network-layer measurements count distinct hops or AS paths as distinct alternatives. It does not identify the submarine infrastructure that could carry those paths. Two paths that appear independent in traceroute may leave a region through the same coast and rely on the same small set of physical directions. Whether network-layer diversity corresponds to physical-corridor diversity is an empirical question.

Cross-layer systems make this comparison possible. Internet Atlas, InterTubes, and iGDB connect observed Internet links to long-haul infrastructure; Nautilus associates IP links with candidate submarine cables; Calypso maps routes using cable layouts, network relationships, and inland connectivity; and Xaminer applies cross-layer maps to infrastructure risk analysis [2]–[7]. These systems produce mappings for individual links or paths. We use such mappings to ask a population-level question: does concentration at the network layer track concentration across feasible physical corridors?

We present CLASP, a Cross-Layer Audit of Service Paths. CLASP normalizes each traceroute into visible, geolocated hop-pair segments. For every country-measurement population, ordered endpoint-AS transitions form the network distribution and feasible pairs of landing regions form the corridor distribution. Both distributions contain the same candidate-bearing segments. Each segment contributes one unit of observation mass, allocated among its candidate corridors by the projection score. We call the Spearman rank association between network and corridor concentration across a reported group of units *Layer Agreement*.

We apply CLASP to 490,911 valid IPv4 traceroutes from 18 public RIPE Atlas measurements. The corpus covers all 13 DNS Roots, Wikipedia, Reddit, Netflix Assets, and two dynamic multi-target measurements used as topology references. The aligned one-hour snapshot yields 3,291,063 atomic segments, of which 2,432,559 are mappable and 124,350 support at least one feasible inter-region corridor. The audit contains 370 country-measurement units: 222 DNS Root, 53 application, and 95 topology-reference units.

The observed Layer Agreement differs between geographic groups. Under projection-score weighting, it is 0.632 among 67 island and archipelagic service-facing units and 0.001 among 202 coastal-mainland units. The group difference remains in the same direction under equal-share and Top-1 allocation. These coefficients describe rank associations within the observed snapshot; the analysis does not establish that geography causes the difference.

The service-facing population is also heterogeneous. Of 275 units, 81 (29.5%) have a broad network distribution and a concentrated corridor distribution, while 158 (57.5%) remain broad at both layers. Other units are concentrated at both layers or expand after projection. A single contraction rate therefore does not represent all country-service populations.

This paper makes three contributions. First, CLASP constructs paired network and corridor distributions from identical traceroute segments. Second, we measure a geographic difference in the observed association between the two layers and report it under three existing candidate-allocation rules. Third, we provide a reproducible audit structure that records measurement IDs, configuration values, mapping states, and pipeline counts, allowing the reported distributions and their limitations to be inspected.

II. BACKGROUND AND RELATED WORK

A. Reading Service Paths at the Network Layer

Users reach services through domain names, anycast targets, and replica deployments. Routing selects the resulting path, and traceroute exposes the hops and AS transitions along it. Path-diversity measurements count these transitions and compare them across vantage points.

Prior work measures the user's view of submarine criticality [8], characterizes anycast deployment [9], [10], and relates cable infrastructure to AS-level structure [11]. These measurements describe network-layer reachability. Our question is whether diversity at that layer corresponds to breadth across feasible physical directions.

B. Mapping Paths onto Submarine Infrastructure

Internet Atlas and InterTubes associate long-haul facilities with observed paths [2], [3]. iGDB integrates facilities, fiber, and network entities into a cross-layer map [4]. Nautilus associates IP links with candidate submarine cables and attaches a confidence score to each assignment [5]. Calypso combines cable layouts, network relationships, and inland connectivity to map routes and defines Route Stress over those assignments [6]. Xaminer applies cross-layer maps to infrastructure analysis under disaster models [7]. Carisimo et al. use intercontinental-link measurements to identify gateway routers [12].

These systems produce an assignment from a link or path to physical infrastructure that could plausibly carry it. Their validation evaluates those assignments using cable failures, targeted traceroutes, operator maps, or expert feedback. CLASP adopts this projection approach; it does not claim to identify ground-truth cable use.

C. From Mapping to Distribution Measurement

An assignment does not by itself describe how observation mass is distributed across physical directions or whether that distribution tracks the one visible at the network layer. CLASP aggregates assignments into two population-level quantities. *Submarine exposure* is the fraction of valid traceroutes containing a feasible inter-region corridor. *Corridor concentration* describes how candidate-bearing segment mass is distributed across corridors. A country can enter the candidate space frequently and distribute observations widely, or enter infrequently with the candidate-bearing observations concentrated on one direction. Section III defines both quantities and preserves their separate denominators.

III. THE PAIRED AUDIT

A. Overview

CLASP builds two distributions over one observation population for each country–measurement unit. It takes traceroutes, probe metadata, IP geolocation and ASN mappings, and cable metadata as inputs. It then extracts segments, projects feasible corridors, and aggregates paired distributions, as shown in Fig. 1.

Exposure and conditional concentration use different denominators. Exposure records whether a valid traceroute contains an inter-region feasible corridor. The paired distributions condition on candidate-bearing segments and describe how their mass is organized.

B. Segments from Service Paths

We group traceroutes by probe country and measured service, retaining the source-side context and the destination selection observed for each measurement. Hop countries remain segment attributes rather than grouping keys.

Each traceroute is normalized into its complete sequence of visible hops with locations, ASNs, and RTTs. Successive visible, geolocated hops form atomic segments. Each segment records endpoint locations, ASNs, countries, positions, RTT difference, and bridged-hop count. Repeated hops and RTT samples retain their trace identity and ordering. When timeouts separate visible hops, the bridged-hop count records the gap.

When both endpoint ASNs are available, their ordered pair forms the network- transition label. A segment missing one endpoint ASN receives an ordered country-pair fallback label. Audit eligibility requires a country-fallback share of at most 30%, at least 30 candidate-bearing segments, 10 probes, and 3 probe ASNs.

C. Feasible Corridors

For each segment endpoint, we retrieve landing stations within 50 km and pair them through cable systems in service on July 1, 2026. Explicit cable path or branch information defines direct landing-point adjacency. When the metadata does not provide this information, the pipeline retains the candidate at the topology resolution supplied by the inventory and records that state.

We group landing stations into regions with maximum pairwise distance 30 km. A *landing-region corridor* is a direction-independent pair of regions connected by at least one feasible cable candidate. Several cable systems and exact landing pairs can support one corridor, while distinct coastal entry–exit patterns remain separate.

A candidate must satisfy geographic, lifecycle, and propagation constraints. Let D be the great-circle distance between its landing points, $v_f = 200$ km/ms the effective propagation speed in fiber, ΔRTT the segment's RTT difference, and $\tau = 5$ ms the configured tolerance. The candidate is propagation-feasible when

$$\Delta RTT + \tau \geq \frac{2D}{v_f}. \quad (1)$$

When ΔRTT is non-positive or inconclusive, the pipeline retains the geographic and lifecycle constraints and flags the segment.

For a segment s with feasible corridor set $\mathcal{C}_s$, the primary analysis allocates mass using the existing projection score:

$$w_{s,c} = \frac{\sigma_{s,c}}{\sum_{c' \in \mathcal{C}_s} \sigma_{s,c'}}, \quad \sum_{c \in \mathcal{C}_s} w_{s,c} = 1, \quad (2)$$

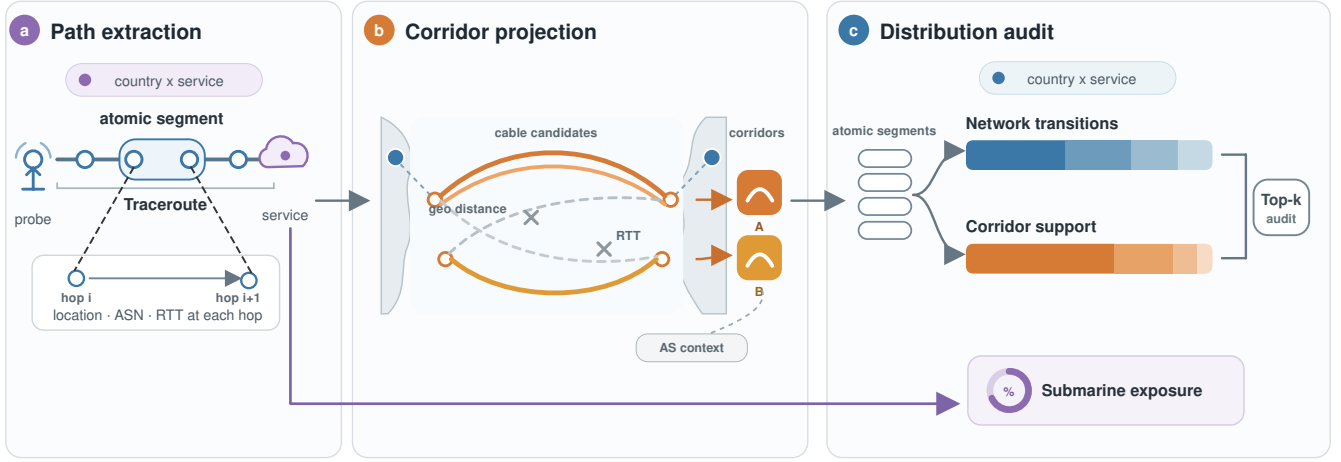


Fig. 1. CLASP constructs paired network-transition and feasible-corridor distributions from candidate-bearing segments in the same observation population.

where $\sigma_{s,c}$ is the candidate support aggregated for corridor c . Every candidate-bearing segment contributes one unit of total corridor mass. The analysis also reports equal-share allocation, $w_{s,c} = 1/|C_s|$, and Top-1 allocation, which assigns the unit to the highest-scoring corridor.

D. Two Distributions over One Population

For an auditable unit u , let S_u be its candidate-bearing segments and $\ell(s)$ the ordered network-transition label for segment s . The network probability of label t is

$$p_u^N(t) = \frac{\sum_{s \in S_u} \mathbf{1}[\ell(s) = t]}{|S_u|}. \quad (3)$$

The corridor distribution sums $w_{s,c}$ from Eq. (2) and normalizes within the same S_u . Both distributions therefore have total mass $|S_u|$ before normalization. Their difference describes how the same candidate-bearing observations are organized at the two layers.

E. Summaries and Layer Agreement

Top-2 share is the sum of the two largest probabilities in a distribution. The effective category count is $\exp[-\sum_i p_i \log p_i]$. Normalized entropy divides Shannon entropy by the logarithm of observed support size. Each quantity is computed within one unit.

Layer Agreement is the Spearman rank association, across a reported set of units, between network Top-2 share and corridor Top-2 share. It describes an observed monotonic association and is not a causal or predictive model. We report the coefficient separately by measurement family, geography, and candidate-allocation rule.

A common 80% Top-2 threshold supplies four descriptive classes: broad at both layers, concentrated at both, network-broad/corridor-concentrated, and network-concentrated/corridor-broad. Continuous Top-2 shifts, effective counts, and normalized entropy accompany this threshold view.

TABLE I
CLASP CONFIGURATION.

Parameter	Value
Endpoint landing catchment	50 km
Landing-region maximum diameter	30 km
Fiber propagation speed	200 km/ms
RTT tolerance	5 ms
Same-city threshold	25 km
Cable lifecycle date	July 1, 2026
Inconclusive RTT	Retained and flagged
Timeout-bridged segment	Retained and flagged
Path scope	Complete visible sequence

IV. DATA

The study uses path and supporting-data versions labeled July 1, 2026. The path observations cover 00:00–01:00 UTC on that date.

A. RIPE Atlas Measurements

The corpus contains 18 public RIPE Atlas IPv4 traceroute measurements. Thirteen cover independently operated anycast DNS Root services; Wikipedia, Reddit, and `assets.nflxext.com` provide three application-facing measurements; measurements 5051 and 5151 use dynamic targets as multi-target UDP and ICMP references. Table II reports the corpus after canonical trace-identity deduplication.

Each record retains the destination address returned by RIPE Atlas. This preserves both the multi-target structure of the two dynamic-target measurements and the destination instances selected by the service measurements during the window.

The three families are not a controlled comparison. The dynamic-target measurements sample many destinations, whereas each service measurement samples the destinations selected for that service. We use the former as topology references and do not attribute family differences to service deployment or target selection.

TABLE II
RIPE ATLAS MEASUREMENT CORPUS.

Group	Target	MSM ID(s)	Raw	Valid
DNS Roots	A–M Root	5009, 5010, 5011, 5012, 5013, 5004, 5014, 5015, 5005, 5016, 5001, 5008, 5006	534,276	317,873
Applications	Wikipedia; Reddit; Netflix Assets	86710103; 176906957; 176517335	115,596	115,596
Topology references	IPv4 UDP; IPv4 ICMP	5051; 5151	57,442	57,442
Total	18 measurements	–	707,314	490,911

B. Supporting Data

RIPE Atlas probe metadata supplies probe country and source ASN [13]. IPinfo Location and ASN databases annotate visible hops [14]. The Submarine Cable Map provides 694 cables and 1,916 landing points, including lifecycle and organizational fields [1]. CAIDA AS Relationships supplies supplementary AS context [15]. A versioned country taxonomy classifies probe countries as island or archipelagic, landlocked, or other coastal.

V. RESULTS

The pipeline retains 490,911 valid traceroutes from 707,314 raw records and produces 3,291,063 atomic segments. Of 2,432,559 mappable segments, 555,119 reach candidate landing points, 171,232 satisfy the corridor constraints, and 124,350 support at least one feasible inter-region corridor. The 370 auditable country–measurement units contain 222 DNS Root, 53 application, and 95 topology-reference units.

A. Inter-Region Candidate Exposure

Inter-region candidates occur infrequently for most service-facing populations. Pooling eligible Root observations within each country by valid- trace count yields median DNS exposure of 5.50% (IQR 1.10–23.04%; mean 17.80%). Median country–measurement exposure is 1.90% for Wikipedia, 1.96% for Reddit, and 1.43% for Netflix Assets. The two dynamic multi- target references reach 23.24% for MSM 5051 and 23.81% for MSM 5151.

Country-matched comparisons retain source country but change the target population. DNS exposure is lower than MSM 5051 in 70 of 77 shared countries, with a median paired difference of (–14.52) percentage points. It is lower than MSM 5151 in 54 of 59 shared countries, with a median difference of (–14.82) points. These comparisons describe the measured populations; they do not identify a deployment or target-selection effect.

B. Layer Agreement within Measurement Families

Pooling all 370 units gives a rank association of 0.47 between network and corridor Top-2 shares. This pooled coefficient includes the separation between measurement families and does not isolate the within-family relationship.

Table IV reports Layer Agreement within each family under the three existing allocation rules. Projection-score weighting gives 0.176 for 219 DNS units, 0.229 for 53 application units,

and 0.012 for 95 topology-reference units. The coefficients describe the ordering of units within each family.

The equal-share primary outputs also show different absolute corridor breadth across the observed families. Median corridor Top-2 share is 72.0% for DNS, 70.3% for applications, and 30.6% for topology references; median effective corridor counts are 3.84, 3.51, and 18.01. Because the target populations differ, these levels are descriptive family summaries rather than a controlled estimate of a service effect.

C. Layer Agreement Differs by Geography

The geographic split separates service-facing populations that are combined in the family-level summaries. Under projection-score weighting, Layer Agreement is 0.632 across 67 island and archipelagic units and 0.001 across 202 coastal-mainland units. The observed difference is 0.631. Landlocked service-facing units are not members of either geographic group in this comparison.

The geographic difference remains in the same direction under all three allocation rules. The association increases as the allocation places more mass on higher-scoring corridors. These results record sensitivity to candidate allocation; without physical-route ground truth, they do not establish that one allocation recovers the true association.

D. Exposure and Conditional Breadth

Geography also separates exposure from conditional corridor breadth in the equal-share outputs. Island and archipelagic countries have median pooled DNS exposure of 49.61% across 9 countries, compared with 7.88% for 54 other coastal countries and 1.70% for 14 landlocked countries.

Conditional breadth follows a different ordering. Island and archipelagic countries have median corridor Top-2 share of 53.3% and 5.58 effective corridors. Other coastal countries reach 75.4% and 3.39. The two represented landlocked countries reach 80.2% and 2.51. Across 38 countries with both pooled exposure and an auditable corridor summary, exposure has Spearman association (–0.29) with corridor Top-2 share and 0.30 with effective corridor count.

E. Contraction Is Not the Only Outcome

Using the paper’s descriptive 80% Top-2 threshold, 81 of 275 service-facing units (29.5%) are network-broad/corridor-concentrated and 158 (57.5%) remain broad at both layers. Eighteen (6.5%) are concentrated at both layers, and 18 (6.5%)

TABLE III
SUPPORTING DATASETS.

Dataset	Version label	Content
RIPE Atlas probe metadata	2026-07-01	Probe country and source ASN
IPinfo Location MMDB	2026-07-01	Hop geolocation
IPinfo ASN MMDB	2026-07-01	IP-AS mapping
Submarine Cable Map	2026-07-01	694 cables; 1,916 landing points
CAIDA AS Relationships	2026-07-01	AS relationships
Country geography taxonomy	2026-07-01	Operational geography classes

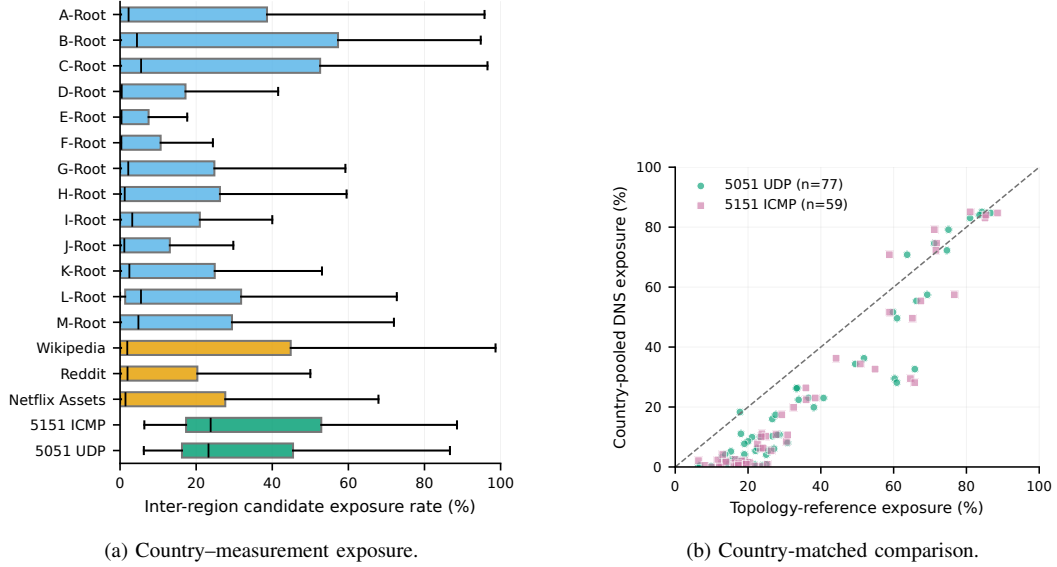


Fig. 2. Inter-region candidate exposure in the observed measurement populations.

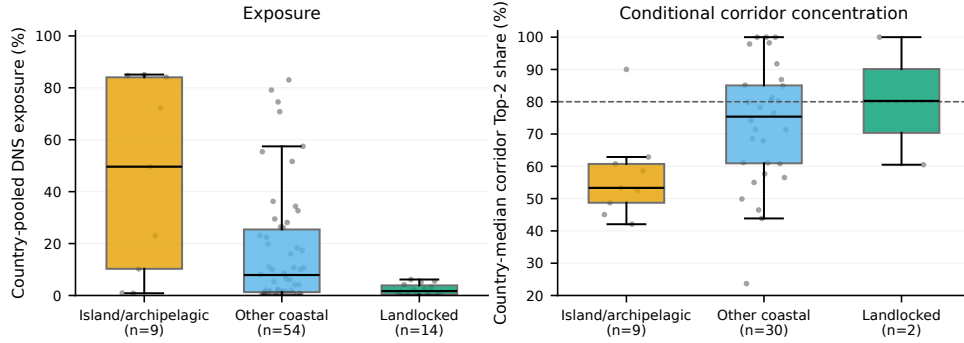


Fig. 3. DNS exposure and equal-share corridor concentration by geography.

TABLE IV
LAYER AGREEMENT BY MEASUREMENT FAMILY.

Family	(n)	Equal	Score	Top-1
DNS Roots	219	0.096	0.176	0.330
Applications	53	0.030	0.229	0.372
Topology references	95	0.139	0.012	0.122

TABLE V
SERVICE-FACING LAYER AGREEMENT BY GEOGRAPHY AND ALLOCATION RULE.

Allocation	Island/arch.	Coastal mainland	Difference
Equal share	0.380	-0.081	0.461
Projection score	0.632	0.001	0.631
Top-1	0.820	0.105	0.715

are network-concentrated/corridor-broad. Four of 95 topology-reference units (4.2%) enter the contraction class; the other 91

remain broad at both layers.

Continuous measures show changes within the threshold

TABLE VI
A-ROOT LANDING-REGION DIAMETER SENSITIVITY.

Diameter	Single-corridor	Median Top-2	Units
10 km	47.4%	53.6%	16
20 km	46.2%	62.8%	15
30 km	54.8%	64.4%	15
40 km	90.1%	81.6%	13
50 km	94.8%	84.9%	13

classes. Corridor Top-2 share increases in 184 of 275 service-facing units (66.9%), and effective category count contracts in 204 (74.2%). Of 239 units with a broad network distribution, 178 (74.5%) move upward in Top-2 share: 81 cross the threshold, while 97 remain broad at both layers with a median increase of 6.3 percentage points.

Three existing cases show different outcomes. **Singapore–Netflix** moves from 32.5% network Top-2 share to 65.0% corridor Top-2 share, while its effective count contracts from 19.86 to 3.41. It narrows without crossing the 80% threshold. **New Zealand–Netflix** moves from 63.6% to 37.6%, with effective count expanding from 5.18 to 6.92. **Singapore–H-Root** moves from 39.6% to 50.1%, and its effective count contracts from 20.67 to 5.99 while normalized entropy changes only slightly. Bounded multi-corridor segments account for 99.9%, 78.1%, and 98.2% of candidate-bearing observations in these cases, respectively.

VI. ROBUSTNESS AND MEASUREMENT SCOPE

A. Candidate Allocation

Table V reports all three allocation rules already computed for the geographic comparison. The island–coastal difference is 0.461 under equal shares, 0.631 under projection-score weighting, and 0.715 under Top-1 allocation. The ordering is stable across these rules, while the coefficient magnitudes are not. The paper therefore reports the allocation rule with each result.

B. Landing-Region Resolution

The existing landing-region-diameter analysis covers the A-Root subset, not all 18 measurements. It varies maximum diameter over 10, 20, 30, 40, and 50 km while holding the 50-km endpoint catchment and 5-ms RTT tolerance fixed. Table VI reports the recorded values. Exact landing-pair and cable candidate sets have mean Jaccard 1.0 on segments shared with the 30-km baseline, but the candidate-bearing segment population and the number of auditable units change with diameter.

The 10–30-km settings retain a finer coastal partition than the 40–50-km settings. Because the existing analysis is limited to A-Root and does not recompute the main geographic Layer Agreement result, it does not establish that the headline geographic difference is invariant to landing-region diameter.

C. Dependence on Geolocation and Candidate Construction

Every corridor projection begins with hop geolocation. A hop placed farther inland can remove a nearby landing candidate; a location shifted toward a coast can add one. The 50-km catchment is the fixed setting used in the reported analysis. No location-perturbation or alternative-geolocation result is available in the existing experiment set, so the effect of residual IPinfo error is not bounded here.

The projection also lacks physical-route ground truth. Geographic, lifecycle, and propagation constraints define feasible candidates rather than observed cable use. Exact landing-pair, cable, and corridor identities remain linked in the pipeline outputs, but no completed manual corridor audit is reported in this draft.

D. Snapshot and Population Scope

All 18 measurements cover 00:00–01:00 UTC on July 1, 2026. Routing, service deployment, and cable availability can change outside this window, so the reported distributions describe one aligned snapshot. RIPE Atlas probe density varies across countries and networks, and all quantities are observation- weighted over participating probes.

The service measurements and dynamic multi-target references have different target populations. Family comparisons describe those observed populations and are not matched estimates of a service-selection or deployment effect. Anycast observations jointly reflect deployed footprint, BGP selection, interconnection, and source-network context.

Cable lifecycle filtering uses the inventory fields available for the measurement date. These fields do not report transient failures or outages, and the cable inventory can be revised. The results therefore measure feasible candidate support under the recorded metadata, not operational redundancy, hazard correlation, rerouting capacity, or repair performance.

VII. DISCUSSION

A. Implications for Diversity Measurement

Layer Agreement separates two questions that a network-layer count can otherwise combine. Network Top-2 share records concentration among observed network transitions. Corridor Top-2 share records concentration among feasible physical directions for the same candidate-bearing segments. In the observed snapshot, the association between these quantities is stronger for island and archipelagic service-facing units than for coastal-mainland units under all three candidate-allocation rules.

This geographic difference is descriptive. One possible explanation is that island paths reach submarine infrastructure through a smaller set of coastal gateways, whereas coastal-mainland paths can include more terrestrial routing. The current measurements do not test that mechanism, and exposure level alone does not establish it.

The heterogeneous threshold classes also limit aggregate interpretations. A broad network distribution projects to a concentrated corridor distribution in 81 service-facing units, but 158 remain broad at both layers and some units expand.

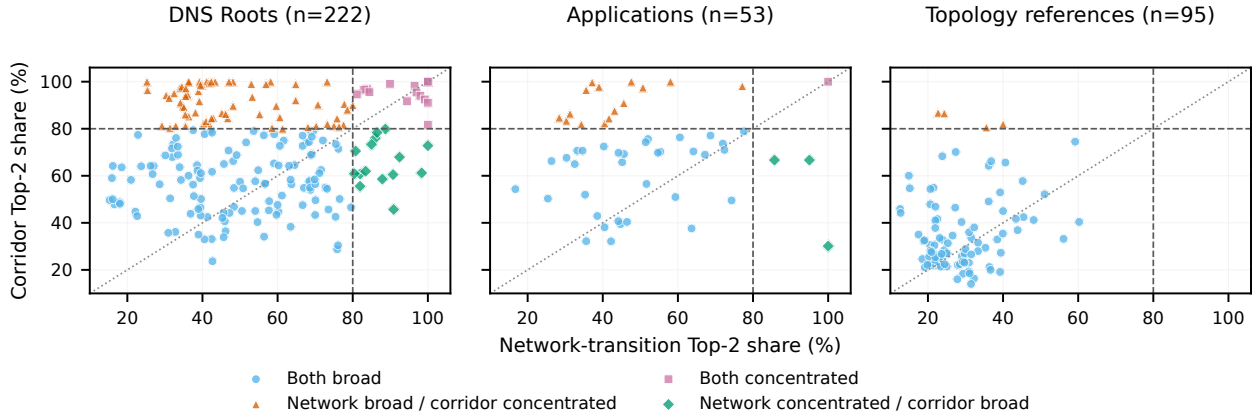


Fig. 4. Equal-share network-transition and corridor Top-2 shares for the 370 auditable units. The 80% lines define the descriptive classes.

Network-layer breadth therefore cannot be converted into a physical- corridor breadth estimate using one universal contraction rate.

B. Operational Scope

Exposure and corridor concentration identify different intervention points, but the measurements do not evaluate interventions. A closer service instance or different interconnection could change how often paths enter the inter- region candidate space. Access through distinct landing regions could change the conditional corridor distribution. These are implications of the two measured quantities, not effects estimated by this study.

Corridor concentration also does not measure cable multiplicity, landing- station redundancy, correlated hazards, available rerouting capacity, or repair processes. These dimensions are required before the reported feasible-corridor distributions can be interpreted as operational resilience.

C. Reproducibility and Ethics

The study uses archived public RIPE Atlas measurements and infrastructure metadata and reports aggregate country-measurement statistics. The pipeline records measurement IDs, configuration values, mapping-resolution states, filtering counts, and the inputs used to produce the tables and figures. The paper’s 80% four-class rule must be distinguished from the repository’s tier-based auxiliary classification when artifacts are released.

The measurements do not target individual users and do not report individual probe addresses. Candidate assignments are feasible hypotheses and should not be interpreted as disclosure of an operator’s actual cable use.

VIII. CONCLUSION

CLASP compares network-transition and feasible-corridor distributions over the same traceroute segments. Across the existing one-hour RIPE Atlas snapshot, projection-score Layer Agreement is 0.632 for 67 island and archipelagic service-facing units and 0.001 for 202 coastal-mainland units; the

observed group difference remains in the same direction under equal-share and Top-1 allocation. The service-facing population is not uniformly contractive: 81 of 275 units are network-broad/corridor-concentrated, while 158 remain broad at both layers. These results show that network-layer diversity and feasible physical-corridor diversity have a geographic and population-specific relationship in the measured snapshot.

IX. USE OF AI DISCLOSURE

OpenAI ChatGPT and Codex assisted with code development, plotting scripts, and language revision. Their contribution was limited to these tasks, and the authors validated the outputs against the source data and analysis results.

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